SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

[PROJECT TITLE]

Detailed Project Report

Submitted by

Submitted by
[Student 1 Name – Roll No.]
[Student 2 Name – Roll No.]
[Student 3 Name – Roll No.]

B.Tech Computer Engineering
Third Year / Semester: V

Faculty: [Faculty Name]
Academic Year: 2026–2027

Mini Project: FSD-MERN Stack

GitHub Repository: [Paste Repository URL]

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1. Introduction

1.1 Background

GUIDELINE – DELETE BEFORE SUBMISSION: *Explain the domain/problem and why the proposed web application is required. Keep this to about 2–3 paragraphs.*

[Write background here.]

1.2 Problem Statement

GUIDELINE – DELETE BEFORE SUBMISSION: *Clearly state who faces the problem, what the problem is, and what improvement is required.*

[Write problem statement here.]

1.3 Proposed Solution

GUIDELINE – DELETE BEFORE SUBMISSION: *Explain what the application does and its main user flow. Mention MERN technologies in relation to the solution.*

[Describe proposed solution here.]

1.4 Objectives and Scope

GUIDELINE – DELETE BEFORE SUBMISSION: *Combine objectives and scope in one section. List 4–6 objectives and briefly state what is included/excluded.*

- [Objective 1]
- [Objective 2]
- [Objective 3]
- [Objective 4]
- [Objective 5]

[Scope of the project]

2. REQUIREMENTS AND TECHNOLOGY STACK

2.1 Users and Functional Requirements

GUIDELINE – DELETE BEFORE SUBMISSION: *List the major functions actually implemented.*

ID	User/Function	Description
FR1	[Function]	[Description]
FR2	[Function]	[Description]
FR3	[Function]	[Description]
FR4	[Function]	[Description]
FR5	[Function]	[Description]

2.2 Non-Functional Requirements

Category	Requirement
Usability	[Requirement]
Performance	[Requirement]
Reliability	[Requirement]
Security	[Requirement]
Maintainability	[Requirement]

2.3 MERN Technology Stack

GUIDELINE – DELETE BEFORE SUBMISSION: *Give one consolidated table instead of separate repeated technology sections.*

Layer	Technology	Purpose
Frontend	React.js	UI and components
Backend	Node.js + Express.js	Server and REST APIs
Database	MongoDB + Mongoose	Data storage and modelling
API/Testing	Postman / Thunder Client	API testing
Version Control	Git + GitHub	Source control
Deployment	[Platform / N.A.]	Hosting, if used

3. SYSTEM DESIGN

3.1 System Architecture

GUIDELINE – DELETE BEFORE SUBMISSION: *Insert one clear diagram showing the actual project architecture and request/response path.*

3.2 Modules and Workflow

Module	Main Functions
[Module 1]	[Functions]
[Module 2]	[Functions]
[Module 3]	[Functions]
[Module 4]	[Functions]

[INSERT FIGURE 3 – APPLICATION WORKFLOW/DATA FLOW HERE]

3.3 Database Design

GUIDELINE – DELETE BEFORE SUBMISSION: *Show the collections and important fields. If there are relationships, show them in the diagram.*

Figure 4 – Database design/schema

Collection	Important Fields	Purpose
[Collection 1]	[fields]	[Purpose]
[Collection 2]	[fields]	[Purpose]
[Collection 3]	[fields]	[Purpose]

3.4 API Design

GUIDELINE – DELETE BEFORE SUBMISSION: *List only the APIs implemented. API execution/testing is covered later in Chapter 5.*

Method	Endpoint	Purpose
POST	/api/[resource]	Create
GET	/api/[resource]	Read all
GET	/api/[resource]/:id	Read one
PUT/PATCH	/api/[resource]/:id	Update
DELETE	/api/[resource]/:id	Delete

4. APPLICATION DESIGN AND IMPLEMENTATION

4.1 User Interface and Navigation

GUIDELINE – DELETE BEFORE SUBMISSION: *Insert 2–3 representative screenshots and explain the main screens, navigation and responsive design. Avoid inserting every page.*

4.2 Frontend Implementation

GUIDELINE – DELETE BEFORE SUBMISSION: *Explain components, state, forms, hooks/routing and API calls actually used. Do not paste full source code.*

[Describe React implementation and important components.]

4.3 Backend and API Implementation

GUIDELINE – DELETE BEFORE SUBMISSION: *Explain Express server, middleware, routes, request handling and response handling.*

[Describe Node.js/Express implementation.]

4.4 Database and CRUD Implementation

GUIDELINE – DELETE BEFORE SUBMISSION: *Explain Mongoose schema/model and how Create, Read, Update and Delete operations are implemented. Avoid repeating the API table from Chapter 3.*

CRUD	Frontend Action	Backend/Database Action
Create	[Form/action]	[POST + create]
Read	[Page/action]	[GET + find]
Update	[Edit/action]	[PUT/PATCH + update]
Delete	[Delete/action]	[DELETE + delete]

4.5 Validation and Error Handling

GUIDELINE – DELETE BEFORE SUBMISSION: *Give 2–3 actual examples of input validation or API/database errors and explain how the application handles them.*

[Describe validation and error handling.]

5. TESTING AND RESULTS

5.1 Testing Approach

GUIDELINE – DELETE BEFORE SUBMISSION: *Mention only the testing performed: API testing, integration/manual testing and database verification. Do not claim automated testing unless implemented.*

[Describe testing approach.]

5.2 Test Cases

GUIDELINE – DELETE BEFORE SUBMISSION: *Use actual inputs and actual results. Mark PASS/FAIL only after execution.*

ID	Test Scenario	Expected Result	Actual Result	Status
TC01	Create record	Record created	[actual]	PASS/FAIL
TC02	View records	Records displayed	[actual]	PASS/FAIL
TC03	Update record	Record updated	[actual]	PASS/FAIL

5.3 API and Database Verification

GUIDELINE – DELETE BEFORE SUBMISSION: *Insert selected evidence only. Use one or two Postman screenshots and one MongoDB screenshot rather than repeating every operation.*

5.4 Final Results

GUIDELINE – DELETE BEFORE SUBMISSION: *State whether the major functional requirements were achieved and tested. Mention deployment status if applicable.*

[Summarize final results.]

6. GITHUB AND DEPLOYMENT

6.1 GitHub Repository

GUIDELINE – DELETE BEFORE SUBMISSION: *Show the important repository structure and briefly explain how the team used Git/GitHub.*

[INSERT FIGURE 8 – GITHUB REPOSITORY HERE]

Figure 8 – GitHub repository

6.2 Deployment Details

GUIDELINE – DELETE BEFORE SUBMISSION: *If deployed, provide correct URLs and platform details. If not deployed, write N.A. Do not invent URLs.*

Item	Details
GitHub Repository	[URL]
Frontend URL	[URL / N.A.]
Backend URL	[URL / N.A.]
Database	[MongoDB Atlas/local]
Deployment Platform	[Platform / N.A.]

7. CHALLENGES AND INDIVIDUAL CONTRIBUTIONS

7.1 Challenges and Solutions

GUIDELINE – DELETE BEFORE SUBMISSION: *List only genuine technical challenges faced during development and the solution used.*

Challenge	Solution	Outcome
[Challenge 1]	[Solution]	[Outcome]
[Challenge 2]	[Solution]	[Outcome]
[Challenge 3]	[Solution]	[Outcome]

7.2 Individual Contributions

GUIDELINE – DELETE BEFORE SUBMISSION: *Record actual contributions. Do not assign work that was not performed.*

Student	Frontend	Backend/API	Database	Testing/Deployment
[Student 1]	[work]	[work]	[work]	[work]
[Student 2]	[work]	[work]	[work]	[work]
[Student 3]	[work]	[work]	[work]	[work]

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

GUIDELINE – DELETE BEFORE SUBMISSION: *Write 1–2 paragraphs summarizing the problem, solution, MERN implementation, database/API/CRUD integration, testing and final outcome.*

[Write conclusion here.]

8.2 Future Scope

GUIDELINE – DELETE BEFORE SUBMISSION: *List realistic features that are not currently implemented.*

References

[1] Aqua Security, "Trivy documentation," https://trivy.dev/.

[2] GitHub, "GitHub Actions documentation," https://docs.github.com/actions.